Corruption, Fairness, and Inequality

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ABSTRCAT

This study examines two prominent arguments for corruption—Uslaner's "inequality trap" thesis, according to which high inequality leads to low trust and thus greater corruption *ad infinitum*—and the unfairness theory. The perception of corruption was measured across 32 countries via the 2006 Role of Government module of the International Social Survey Program (ISSP). The findings indicate that, in line with the "inequality trap" thesis, countries with higher levels of income inequality evince a higher degree of perception of corruption and, in correspondence with the unfairness thesis, that people who believe that public officials treat them fairly are less likely to perceive corruption to exist. Combining these two explanations, we employed a multilevel model to examine whether the negative correlation between fair treatment and perceptions of corruption vary in accordance with the level of country income inequality. The results indicate that the link between these variables is weaker in less equal countries than in more egalitarian countries. In unequal countries, fairness doesn't matter quite so much for corruption perceptions.

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Introduction

What drives perceptions of corruption? Ordinary people find corruption morally objectionable. What irks them most—and what makes them more likely to perceive high levels of corruption, we argue, are perceptions that elites grow rich from dishonesty and that they gain special treatment not available to ordinary people. When people see the rich getting richer and believe that they got their wealth unfairly, they are most likely to perceive high levels of corruption.

We examine two prominent arguments about the roots of corruption—and how people view it. The first is Uslaner's (2008) "inequality trap" thesis. Corruption, on his argument, stems from high inequality and low trust in other people. High inequality leads to low trust and thus to high corruption, which in turn leads to more inequality. The second, most prominently associated with Rothstein (2011, 103-110) and You and Khagram (2005), is the claim that corruption rests upon a foundation of unfairness.

Perceptions of fair treatment lie at the heart of perceptions of grand corruption– but only in relatively equal societies. In unequal societies, people don't expect fair treatment. They don't see corruption as acceptable, but as inevitable and beyond their control. Inequality demoralizes people and leads them to believe that they are not masters of their own fate. In societies with more equitable distributions of resources, people have greater expectations of fairness. They presume that unfair treatment is more than simply wrong–it represents a violation of the integrity of the governmental system.

We combine the arguments of Uslaner's "inequality trap" with the fairness thesis of Rothstein and others. We shall show that both people who do not trust others

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or government are more likely to perceive grand corruption in their countries. We also show that perceptions of fair treatment by officials and needing connections with public officials to obtain favorable outcomes shape perceptions of corruption. These effects are exaggerated in the countries that are more equal and smaller changes in perceived fairness lead to much lower expectations of corruption. We find, perhaps counterintuitively, the most unequal countries, fairness doesn't matter quite so much for corruption perceptions.

We support our claims by estimating multilevel models of corruption perceptions in the 2006 Role of Government module of the International Social Survey Program (ISSP) across 32 countries in Europe, Asia, Latin America, and South Africa. The wide range of countries in the ISSP module assures us that our arguments are not specific to one region or set of countries.

Corruption, Inequality, Trust, and Fairness

There is a widely accepted argument linking corruption and inequality. Glaeser, Scheinkman, and Schleifer (2003, 200; see also You, in press, 45-46) argue:

> ...inequality…enables the rich to subvert the political, regulatory, and legal institutions of society for their own benefit. If one person is sufficiently richer than another, and courts are corruptible, then the legal system will favor the rich, not the just. Likewise, if political and regulatory institutions can be moved by wealth or influence, they will favor the established, not the efficient. This in turn leads the initially well situated to pursue socially harmful acts, recognizing that the legal, political, and regulatory systems will not hold them accountable. Inequality can encourage institutional subversion in two distinct ways. First, the havenots can redistribute from the haves through violence, the

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political process, or other means. Such Robin Hood redistribution jeopardizes property rights, and deters investment by the rich.

Similarly, You and Kaghram (2005, 138) argue: "The rich, as interest groups, firms, or individuals may use bribery or connections to influence law-implementing processes (*bureaucratic corruption*) and to buy favorable interpretations of the law (*judicial corruption*)." Inequality breeds corruption by: (1) leading ordinary citizens to see the system as stacked against them (Uslaner, 2002, 181-183); (2) creating a sense of dependency of ordinary citizens and a sense of pessimism for the future, which in turn undermines the moral dictates of treating your neighbors honestly; and (3) distorting the key institutions of fairness in society, the courts, which ordinary citizens see as their protectors against evil-doers, especially those with more influence than they have (see also Glaeser, Scheinkman, and Schleifer, 2003; You and Khagram, 2005).

Corruption gives some people advantages that others don't have. Corruption transfers resources from the mass public to the elites–and generally from the poor to the rich (Tanzi, 1998). It acts as an extra tax on citizens, leaving less money for public expenditures (Mauro, 1997, 7). Corrupt governments have less money to spend on their own projects, pushing down the salaries of public employees. In turn, these lower-level staffers will be more likely to extort funds from the public purse. Government employees in corrupt societies will thus spend more time lining their own pockets than serving the public. With less money in the public purse, key services won't be provided to those who need it most—the poor. The wealthy can find ways to obtain these services, but the poor may face demands for bribes they can't afford—or even have to go without (Uslaner, 2011).

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While the linkage between inequality and trust is widely acknowledged, empirical support for a direct connection is weaker. Uslaner (2008, 46-48) reported a very modest relationship (r2 = .082) across 85 countries, although You and Khagram (2005, 144-48) found a stronger relationship. Uslaner (2008, 63-73) found much greater support for an indirect effect using aggregate data: Inequality leads to lower levels of interpersonal trust, which in turn gives rise to more corruption. Corruption then reinforces inequality—and promotes more in-group trust at the expense of trust in people unlike yourself—and ultimately more corruption in a vicious circle. As corruption robs the public purse, it will lead to less trust in government as well as to trust in people (Chang and Chu, 2006; Lavallee, Razafindrakoto, and Roubaud, 2008; Uslaner, 2011). Just as high levels of corruption should make people see less trust in both other people and government, people with low trust (by either measure) should be more likely to perceive high levels of corruption. The ISSP does not have sufficient variables to permit estimation of a simultaneous equation model, so we focus on how each type of trust shapes corruption perceptions.

We expect to find a direct linkage between inequality and perceptions of corruption. Even as aggregate level support for this connection is not strong, perceptions of corruption in survey data are more closely linked to inequality (Uslaner, 2008, chs. 4-8; You and Khagram, 2005). You (2006) sees *fair* institutions as the foundation of good (honest) government, while Rothstein (2011) and Kurer (2003) argue that *partiality* in the implementation of public laws and public policies lies at the heart of corruption. We don't see corruption as being defined by impartiality—after all, lots of things in life and in politics are unfair without necessarily being corrupt. We agree that when people believe that they face unfair treatment by government officials, they may be particularly likely to think that better

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treatment for those with more resources reflects corruption. People are especially likely to make this argument *when they have a reason to expect fair treatment*. When inequality is high, people may have lower expectations of impartiality. And when people have long had to put up with an unresponsive bureaucracy, as in the former Communist states, fair treatment may not shape perceptions of corruption as much as they would in other countries.

We focus on perceptions of "grand corruption," malfeasance that involves large stakes, mostly payments by businesspeople to politicians (or to other businesspeople). We don't consider "petty corruption," which involves small payments to the police (to avoid speeding fines), to doctors (for faster service), to teachers (for better grades or to administrators for admission to schools), and to other low-level public servants. These payments are still considered morally wrong, but they help people get around bureaucratic systems that are broken—you pay a small bribe to the police officer or doctor so you don't have to spend an entire day in traffic court or in the waiting room. Petty corruption doesn't involve large sums of money, so it will not make the rich even wealthier as does grand corruption. People associate grand corruption with inequality. They don't make this connection with petty corruption, mostly because the sums are so small (from data on Romania and Africa in Uslaner, 2008, chs. 5, 7).

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Method

Individual-level data

We employ data from the 2006 Role of Government module of the International Social Survey Program (ISSP; Skjåk 2010).[1] The ISSP 2006 is the only survey with data from Western and non-Western nations that includes questions on corruption and perceptions of fairness as well as trust. We analyze a pooled sample of 32 countries: Australia, Canada, Chile, Croatia, Czech Republic, Denmark, Dominican Republic, Finland, France, Germany, Hungary, Ireland, Israel, Japan, Latvia, the Netherlands, New Zealand, Norway, Philippines, Poland, Portugal, Russia, Slovenia, South Africa, South Korea, Spain, Sweden, Switzerland, the United Kingdom, the United States, Uruguay and Venezuela. The number of respondents varies from 930 in Britain to 2,939 in South Africa. The overall pooled sample is 36,692.[2]

Measures

The dependent variable *perception of grand corruption* was measured by two items: "In your opinion, about how many politicians in [r's country] are involved in corruption?" and "And in your opinion, about how many public officials in [r's country] are involved in corruption?" These were measured on a 1-5 scale and are strongly correlated across the pools sample (r = .726).[3] The levels of perceived corruption vary from as low as 1.42 in Denmark to a high of 3.54 in Venezuela. Descriptive data of the polled sample can be found in Appendix 1.

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[1] ZA4700: International Social Survey Programme 2006 (Role of Government IV). Details regarding data collection, sampling, and response rates can be found on the ISSP website: http://www.issp.org.

[2] The multilevel analysis was verified on a balanced sample, so each state's weight from the pooled sample is 3.1%.

[3] The correlations vary from as low as .49 in South Korea to as high as .80 in South Africa.

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Our principal independent variable—fair treatment by public officials—was measured separately via two questions. The first—"In your opinion, how often do public officials deal fairly with people like you?"—relates to citizen perception of *fair treatment*. The second—"Do you think that the treatment people get from public officials in [r's country] depends on who they know?"—relates to citizen perception of *contact*. Higher scores indicate agreements with these two statements.[4] We also include measures of *interpersonal trust* ("There are only a few people I can trust completely") and *political trust* ("People we elect as MPs try to keep the promises they have made during the election"), and respondents' level of education.

Perceptions of unfair treatment should lead people to see more corruption. So should the need for connections. Both are related to perceptions of inequality. People who receive better treatment than others and who have connections benefit have advantages that others do not have. And we expect that people will see this "extra" influence as reflecting corruption.

Uslaner (2008, 2011) has shown that both social and political trust is linked to corruption. In both cases the direction of causality is complex—and likely to go both ways. Without good instruments, we focus on how trust leads to perceptions of greater honesty. When people trust others, they are prone to see mostly the good side of society (Uslaner, 2002, ch. 2) and thus should see less corruption. When we believe that politicians keep their promises, we see them looking out for our interests, not just their own.

Among demographic variables, we only see one that should matter: education. Educated people have less need to rely upon corrupt leaders for their livelihood—and thus should be less forgiving of such malfeasance. Botero, Pontero, and Shleifer

[4] The relatively week correlation (r =.-287) between these two items indicates that they reflect different dimensions of fair treatment by public officials.

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(2012) argue that more highly educated people are more likely to protest against corruption, even in non-democratic states. And there is an indirect effect as well: Education is one of the strongest predictors of interpersonal trust (Uslaner, 2002, ch. 4), which in turn is strongly connected to lower levels of grand corruption.

Finally, since we have argued that people in formerly Communist countries are more likely to perceive unfairness—and to have low levels of trust and to perceive higher levels of corruption (Uslaner, 2008, chs. 3-4). Income inequality has historically been lower in the transition countries, but it has risen sharply in almost every post-Communist society since the fall of the authoritarian regimes in 1989 and succeeding years. So we include a dummy variable for transition countries in our analysis to control for the distinctive effects in these nations.

Country-level data

Our contextual variable is *income inequality*—measured by the Standardized World Income Inequality Database (Solt 2009). Standardizing several data sources, this index maximizes the comparability of income inequality across countries.[5] The SWIID index is the most comprehensive measure of inequality across nations. In addition, the SWIID has post tax measures so it better capture the policy aspect of the redistribution. The index of net income inequality ranges from as high as 68 in South Africa to a low of 23 in Denmark. Since corruption is widespread in former Communist countries (Ledeneva, 2005; Uslaner, 2008, ch. 3), we employ a dummy variable for the seven post-communist countries in the sample for their effect on perceptions of corruption.

[5] See: http://myweb.uiowa.edu/fsolt/swiid/swiid.html.

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Overview of the analysis

Our analytical strategy was based on a multilevel approach that allows the effect contextual variables exert on dependent variables on the individual level to be assessed (Hox 2010). We utilized multilevel regression models to analyze how dissimilar national contexts affect the relationship between fair treatment and perception of grand corruption at the individual level. This allows us to examine how the country-level variable variation corresponds to differences in the intercepts and slopes. We use the interclass correlation coefficient (ICC) to estimate the degree to which the variance in perception of corruption is explained at the country level (Hox 2010). National contexts account for 27 percent of the variation in perceptions of corruption.

We recognize that there is potential endogeniety between perceptions of corruption and fairness. We do not estimate instruments for fairness because the ISSP has a limited range of variables, none of which appears to be a suitable instrument for unfairness.

Results

We present the multilevel regression models regarding the perception of grand corruption in Table 1. Model 1 focuses on the individual-level variables. As the correlation between the individual level variables being lower than .33, no multicollinearity exists. More highly educated people see greater levels of grand corruption. People with lower levels of interpersonal trust and confidence in government are also less likely to perceive high levels of grand corruption. The controls for education and trust do not weaken the ties between fair treatment, on the one hand, and contact, on the other hand, with corruption perceptions. The negative

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relation between fair treatment and perception of corruption and the positive link between contact and corruption —obtains even after controlling for education and trust. Fair treatment and contact are important in explaining perceptions of grand corruption even controlling for social and political trust, education, and the country level variables.

We find stronger impacts in transition countries. Respondents in former Communist countries have corruption perceptions that are 1.74 units greater than others—a substantial effect on this five point scale. South Africans (living in the most unequal country in our sample) are .95 more likely to perceive grand corruption than are Danes. Fair treatment and relying on contacts are the individual-level variables with the greatest impact: People seeing unfair treatment are .4 more likely to perceive grand corruption, while those who see the rely on contacts are .48 more likely to perceive dishonesty. The effects of other individual-level variables are more modest: .29 for political trust, .13 for generalized (interpersonal) trust, and .11 for education. The country level variables have the strongest effects, but individual-level factors matter, especially fair treatment and contact.

Table 1 and Figure 1 around here

In Model 2 we examine the effect of the country-level variables upon perceptions of corruption. This model has two components. The first element examines how mean perceptions of corruption vary according to the countries' inequality.[6] The positive coefficient indicates that perceptions of corruption are higher where there is more income inequality. Figure 1 illustrates this by plotting countries' mean perceptions of corruption and inequality. In Denmark, the most equal country in our sample, the mean of perception of corruption is just 1.42. In the most unequal country we

[6] This is the direct effect (intercept).

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consider, South Africa, the perception of corruption is almost twice as great (2.57 on the five-point scale).

In Model 1 there is a negative strong relation between fair treatment and perceptions of corruption in the pooled sample. These relations vary across the countries from a strong negative link (β=-.197) in South Korea to an insignificant one (β=.015) in the Philippines. In Model 2 we consider whether these differences can be attributed to income inequality by adding an interaction between the country level and individual level variables.[7] The interaction effect tells us whether economic inequality shapes the relationship between fair treatment and corruption perceptions.[8] The positive coefficient indicates that the slopes for countries with higher levels of inequality are uniformly higher than those for nations with more equitable income distributions.[9] In Figure 2 we see that these slopes (the upper line, which is the upper quartile of inequality) are less steep then the lower one.[10] The impact of fair treatment on corruption perceptions is much greater in more equal countries, where people are far more likely to expect fair treatment. When people expect fair treatment from the government and do not receive it, they attribute this to corrupt officials. Where corruption is high, even people who believe that have been treated fairly perceive high levels of corruption. Almost everyone in Denmark (81 percent of respondents) believes that they are treated fairly by officials, while fewer than a third (28 percent) of South Africans agree that they are treated fairly.

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[7] Random slopes model.

[8] Grand mean centering was used. Restricted maximum likelihood was used for estimating, which is robust for such number of clusters (N=32) with normal distribution of the outcome variable

[9] The difference between the slopes is significant ($p < .001$). Replicating the model using an alternative measure for income inequality (uneven economic development indicators from the Failed State Index) does not alter the result.

[10] Omitting possible outliers like Venezuela, Russia or South Africa did not alter the results.

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Fewer than two percent of Danish respondents believe that they are treated unfairly and almost half of those in the middle category (3 on the 5 point scale) also perceive high levels of corruption. In South Africa, perceptions of corruption are widespread (at 71 percent) even among the 20 percent who say that public officials are fair. People in highly equal and highly unequal countries thus have different perceptions of corruption.

Figure 2 around here

We also found (details available upon request) that similar effects obtain with respect to the contact variable. Perceptions of fair treatment (and contact) matter *less* in highly unequal countries and these effects don't vary much across levels of inequality. In the most equal countries, the slopes (impacts) are striper and small increments in perceptions of fair treatment matter make people much less likely to see grand corruption.

Inequality matters both directly and indirectly. Where the distribution of income is skewed towards the top, people are likely to believe that the rich gained their wealth corruptly rather than by honest hard work. Becoming inured to corruption, they accept its inevitability and their powerlessness to curb it (Stoyanov et al., 2000, 35; Uslaner, 2008: 41-45). Perceptions of fairness matter less—if at all—when people have little reason to expect equitable treatment, constituting an important factor when people expect their leaders to be unbiased. Rothstein and others emphasize the importance of institutional fairness in boosting honest society. Where there are high levels of income inequality fair treatment matter less for reducing perceptions of corruption.

Conclusion

Corruption is difficult to see directly. "Well designed" corruption should not be visible—and few leaders would brag about being corrupt, or even admit to it.

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However, ordinary people do perceive levels of corruption and their evaluations are generally quite close to those of elites (Uslaner, 2008, 14-15).

We have found support for Rothstein's argument on fairness. Perceptions of equitable treatment have strong effects on seeing high levels of corruption. We have also found support for Uslaner's "inequality trap" thesis. Both perceptions of corruption and of fairness are shaped by inequality. And greater trust—both interpersonal and political—also make people less likely to see high levels of grand corruption.

When people don't trust their fellow citizens and especially where they believe that their leaders do not keep their promises, they are also more likely to perceive greater corruption. The individual factors most strongly linked to corruption perceptions are believing that you are being treated unfairly and that you need connections to get things done. People who feel left out seem likely to blame their difficulties on corruption rather than assessing responsibility on their own faults.

You don't need to report actual experiences with corruption ("petty corruption") to believe that those at the top got there by being dishonest (cf. Abramo, 2005 with Uslaner, 2008, 14-15). Where you find low-level (petty) corruption, you almost certainly will find high levels of grand corruption (Uslaner, 2008, 16).

Our central result is that people see corruption as rooted in unequal treatment. Corruption is not just legally wrong. It gives privileges to some, those already at the top, and discriminates against others, people with fewer resources. Even when corruption is not widespread and countries are relatively equal, people see unfair treatment by government officials as reflecting a corrupt regime. This should give us some pause in evaluating efforts to reduce corruption without tackling inequality and unfair treatment of ordinary people.

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References

Abramo, Claudio Weber. 2005. "How Far Go Perceptions?" Brasilia: TransparenciaBrasil, at http://www.transparencia.org.br/docs/HowFar.pdf.

Botero, Juan, Alejandro Ponce, and Andrei Shleifer. 2012. "Education and the Quality of Government." NBER Working Paper, available at www/nber.org/papers/w18119

Chang, Eric C.C. and Yun-han Chu. 2006. "Corruption and Trust: Exceptionalism in Asian Democracies?" Journal of Politics, 68:259-271.

Glaeser, Edward L., Rafael La Porta, Florencio Lopez-de-Silanes, and Andrei Shleifer. 2004. "Do Institutions Cause Growth?", National Bureau of Economic Research Working Paper 10568, at http://papers.nber.org/papers/w10568.pdf.

Hox, Joop. J. 2010. Multilevel analysis: Techniques and applications (2nd ed.). New York: Routledge.

Kurer, Oskar. 2003. "Corruption: An Alternative Approach to Its Definition and Measurement," Political Studies, 53:222-239.

Ledeneva, Alena. 2004. "The Genealogy of *Krugovaya Poruka:* Forced Trust as a Feature of Russian Political Culture." In Ivana Markova, ed., Trust and Democratic Transition in Post-Communist Europe. Oxford: Oxford University Press.

Lavellee, Emmanuelle, Mireille Razafindrakoto, and Francois Roubard. 2008. "Corruption and Trust in Political Institutions in Sub-Saharan Africa," Development Institutions and Analyses de Long Term," at http://ideas.repec.org/p/dia/wpaper/dt200807.html

Electronic copy available at: https://ssrn.com/abstract=2460399

Mauro, Paolo 1997. "Why Worry About Corruption?" Washington: International Monetary Fund

Rothstein, Bo. 2011. The Quality of Government. Chicago: University of Chicago Press.

Skjåk, K., Kalgraff. 2010. The international social survey program: Annual cross-national social surveys since 1985. In J. Harkness, M. Braun, B. Edwards, T. P. Johnson, L. E. Lyberg, P. P. Mohler, . T. W. Smith (Eds.), Survey methods in multicultural, multinational, and multiregional contexts (pp. 497-506).

Solt, Fredrick. 2009. Standardizing the world income inequality database. Social Science Quarterly, *90*(2), 231-242.

Stoyanov, Alexander, Maragarita Pavlikianova, Andrej Nontchev, and Galja Krasteva. 2000. "Bulgaria: Political and Economic Crisis; Democratic Consolidation." In David S. Mason and James L. Kluegel, eds., Marketing Democracy. Lanham, MD: Rowman and Littlefield.

Tanzi, Vito. "Corruption around the world: Causes, consequences, scope, and cures." Staff Papers-International Monetary Fund (1998): 559-594.

Uslaner, Eric M. 2002. The Moral Foundations of Trust. New York: Cambridge University Press.

_____. 2008. Corruption, Inequality, and the Rule of Law. New York: Cambridge University Press.

_____. 2011. "Corruption, the Inequality Trap, and Trust in Government." In Sonja Zmerli and Marc Hooghe, eds., Political Trust: Why Context Matters.

You, Jong-sung. (In press). *Democracy, Inequality, and Corruption: Korea, Taiwan, and the Philippines Compared*. New York: Cambridge University Press.

Electronic copy available at: https://ssrn.com/abstract=2460399

You, Jong-sung. 2006. "Corruption as Injustice." Presented at the Annual Meeting of the *Midwest Political Science Association*, Chicago, IL, April.

You, Jong-sung and Sanjeev Khagram. 2005. "A Comparative Study of Inequality and Corruption." American Sociological Review 70:136-157

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Table 1. Multilevel regression models for percpetions of grand corruption

	1	2
Individual level variables		
Education	.022(.005)***	.022(.005)***
Interpersonal trust	-.033(.004)***	-.033(.004)***
Political trust	-.073(.007)***	-.073(.007)***
Fair treatment	-.101(.008)***	-.102(.006)***
Contact	.119(.011)***	.119(.011)***
Country level variables		
Post-Communist countries		.436(.070)***
Income inequality		.021(.004)***
Fair treatment*income inequality		.003(.000)***
Individual level variance	.440	.440
Country level variance	.142	.085
Slope variance	.173	.105

Notes: Robust Standard error in parentheses. ***<.001 level. *Source*: ISSP 2006. N= 36692 countries = 32

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Figure 1. Perception of corruption and income inequality plot

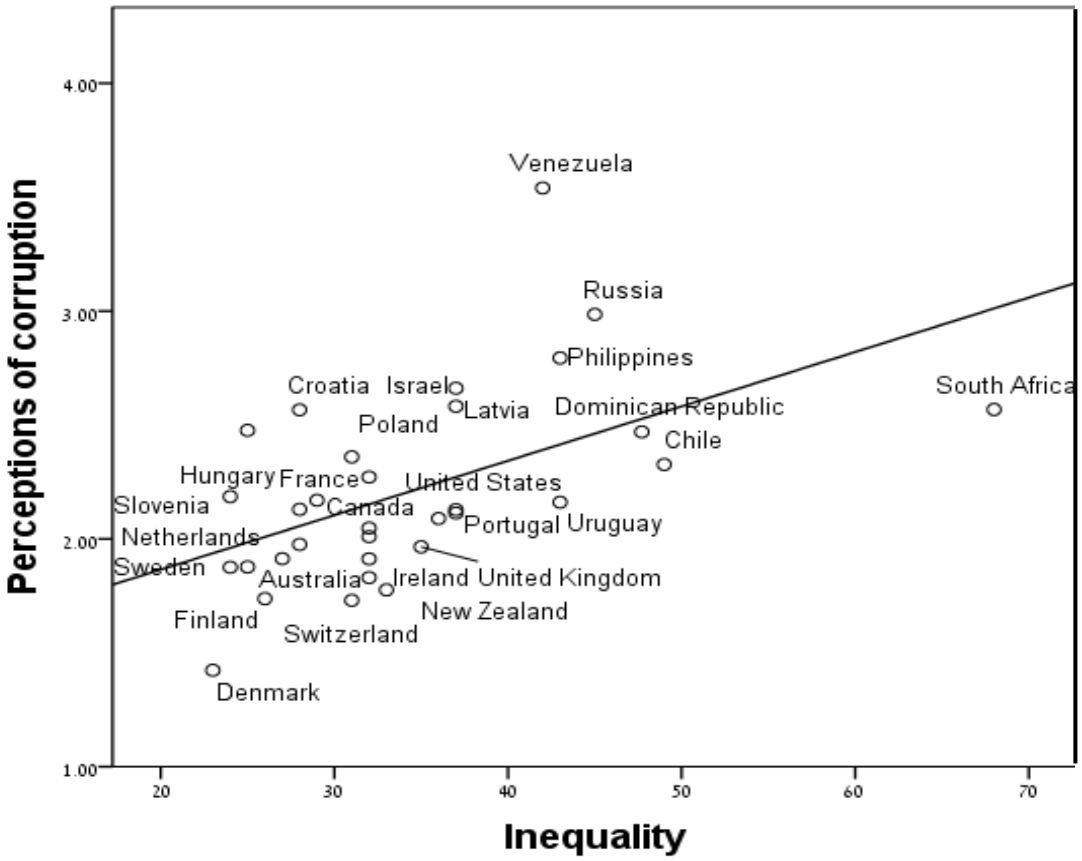

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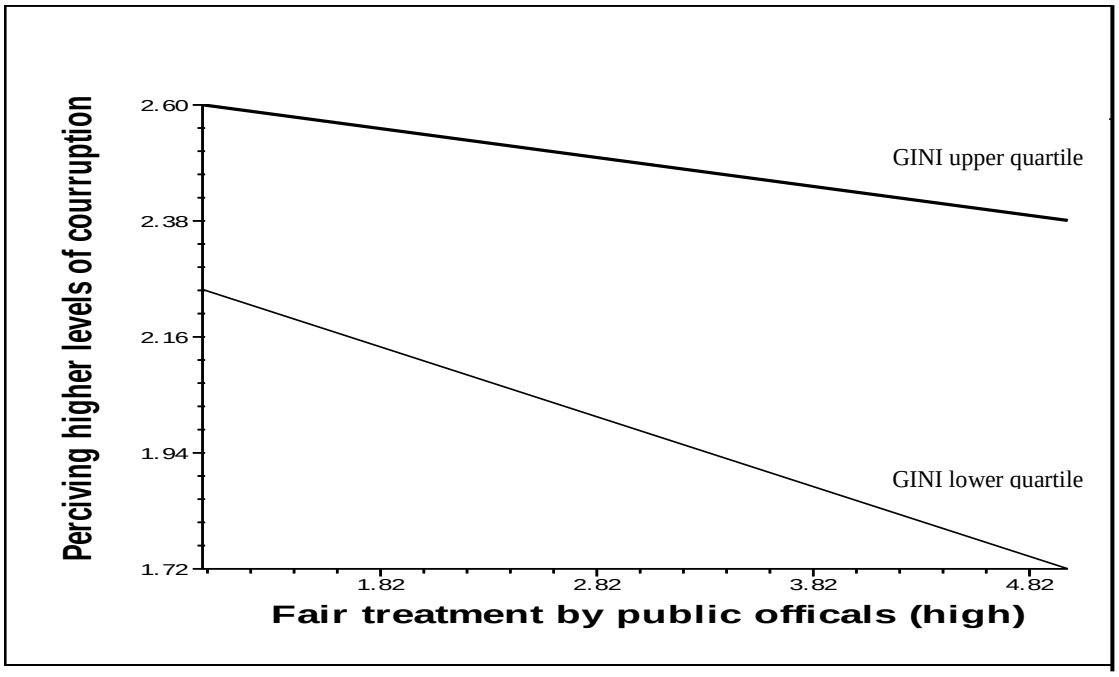
Figure 2. Slope diffrences the relation between fair treatment and perception of corruption under the affects of income inequality (Model 2)



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Appendix 1. Descriptive individual and country level data

	Minimum	Maximum	Mean (SD)
Individual level variables			
Percpetions of grand corruption	1	5	3.26 (1.00)
Education	1	5	2.64 (1.57)
Interpersonal trust	1	5	2.12 (1.08)
Political trust	1	5	2.46 (1.10)
Fair treatment	1	5	3.15 (1.24)
Contact	1	4	3.12 (.81)
Country level variables			
Income inequality	23	68	34.33 (9.36)

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